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22 May 1981
SUPERSEDING
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4 January 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 7895

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, miniature, high Mu, metal-shell amplifier

Outline - - - - - - - - 4-4 (EIA)
Base - - - - - - - - E5-65
Envelope - - - - - - - - MT4
Cathode - - - - - - - - Coated unipotential
Base connections:

Pin No. - - -	1	2	3	4	5	6	7	8	9	10	12
Element - - -	int con	a	int con	g	int con	int con	int con	k	int con	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ebb	Eb	Ec	Ehk	Rk	Rg	Ik	Ic	Pp	TE(shell)	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.9	330	110	+2,-55	100	---	Note 1	15	2.0	1.0	150	100,000
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	110	0	Note 2	150	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current	3		0.4	II	If	125	145	mA
1336	Heater-cathode leakage	3		0.4	II	Ihk	---	3	μAdc
1266	Total grid current	3	Eb = 150 Vdc; Rg = 0.5 Meg; Ecc = -1.7 Vdc; Rk = 0	0.4	II	Ic	---	-0.1	μAdc
1256	Electrode current (1) (anode)	3		0.4	II	Ib	5.5	8.8	mAdc
1306	Transconductance (1)	3		0.4	II	Sm	7,900	10,900	μmhos
1256	Electrode current (2) (anode)		Ec = -5.0 Vdc; Rk = 0	0.4	II	Ib	---	50	μAdc
1201	Short and discontinuity detection	3,5		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		g-all a-all	2.5	II	R R	5,000 10,000	--- ---	Meg Meg
2201	Noise and microphonics	6	Rg = 51,000 ohms; Rp = 2,000 ohms; Ecc = 1.0 V (rms)	2.5	II	---	---	---	--
1306	Transconductance (2)		Ef = 5.7 V	2.5	II	ΔSm / Ef	---	15	%
1316	Amplification factor			6.5	S3	Mu	54	74	---
1231	Emission at reduced filament voltage	7	Ef = 5.5 V; Eb = 90 Vdc; Rk = 0; Ec = 3.5 Vdc	6.5	S3	Is	6	---	mA
1331	Direct-interelectrode capacitance			6.5	Code E	Cin Cout Cbk Cgp Cpk	3.4 1.3 1.0 0.8 0.16	5.0 2.1 1.6 1.0 0.28	pF pF pF pF pF
1031	Sweep-frequency vibration (1)	9	F = 3 to 6 kHz F = 6 to 15 kHz Rp = 2,000 ohms; 1 G; t = 7 seconds, position X only	6.5	Code E	ep ep	--- ---	80 700	mv mv
1031	Sweep-frequency vibration (2)		Rp = 2,000 ohms; 1 G; F = 0.05 to 3 kHz; t = 30 seconds per octave; position X only	6.5	Note 8	Ep	---	35	mVac
1041	Shock	10	Ec = -1.7 Vdc; Rk = 0; Ebk = +100 Vdc; 1,000 G	---	Note 8	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1031	Vibration-fatigue		Eb = Ec = 0; Rk = 0; f = 60 Hz; t = 48 hours; position x only	6.5	Note 6	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1306	Transconductance (1) (change of individual tubes)			---	---	ΔSm t	---	10	%
1266	Total grid current		Eb = 150 Vdc; Rg = 0.5 Meg; Ecc = -1.7 Vdc; Rk = 0	---	---	Ic	---	-0.2	μAdc
1336	Heater-cathode leakage			---	---	Ihk	---	5	μAdc
1031	Sweep-frequency vibration (1)	9	f = 3 to 6 kHz	---	---	ep	---	100	mv
			f = 6 to 15 kHz	---	---	ep	---	1,000	mv
			Rp = 2,000 ohms; 10; t = 7 seconds, position x only;						
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 8.5 V; 1 min "on", 2 min "off"; Ehk = -100 Vdc; Eb = 0; Rk = 0;	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	5	μAdc
1516	Stability life		Ebb = 250 Vdc; Ecc = -1.6 Vdc; Ehk = +100 Vdc; Rg = 0.5 Meg; Rb = 15,000 ohms	---	---	---	---
---	Stability life-test end point:						
1306	Transconductance (1) (change of individual tubes)			ΔSm t	---	10	%
1501	Intermittent life	4	Eb = 150 Vdc; Rg = 0.5 Meg; Ehk = +100 Vdc; Ecc = -1.6 Vdc; Rk = 0; TE = 150°C (min)	---	---	---	---
---	Intermittent life test end points (500 hours):	11					
1266	Total grid current		Eb = 150 Vdc; Rg = 0.5 Meg; Ecc = -1.7 Vdc; Rk = 0	Ic	---	-0.2	μAdc
1301	Heater current			If	125	147	mA
1306	Transconductance (1) change of individual tubes)			ΔSm t	---	15	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	5	μAdc
1211	Insulation of electrodes		g = all	R	1,000	---	Meg
			a = all	R	1,000	---	Meg
1306	Transconductance (1) average change			Avg ΔSm t	---	10	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life test end points (1,000 hours):	11					
1266	Total grid current		Eb = 150 Vdc; Rg = 0.5 Meg; Ecc = -1.7 Vdc; Rk = 0	Ic	---	-0.2	μAdc
1301	Heater current			If	125	149	mA
1306	Transconductance (1) change of individual tubes)			ΔSm 1	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	5	μAdc
1711	Insulation of electrodes		g-all	R	1,000	---	Meg
			a-all	R	1,000	---	Meg
1511	Cathode interface life		Ef = 6.9 V	Ri	---	25	Ohms

NOTES:

1. Maximum grid-circuit resistance for operation at metal-shell temperature up to 150°C:

   For fixed-bias operation: 0.5 Meg
   For cathode-bias operation: 1.0 Meg

2. The reference point for heater-cathode potential shall be the positive terminal of the cathode resistor.

3. This test shall be performed at the conclusion of the holding period.

4. Shell temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), or thermocouple elements welded to shell will be satisfied if a tube under test (TUT) having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

5. The TUT shall be tapped with a hand tapper consisting of a Bakelite rod 1/8-inch in diameter and 7-inches in length with a rubber tip 1-inch long. The rubber tip consists of gum tubing 1/8-inch ID and 3/32-inch wall thickness with a Durometer rating of 35 ±5 Shore A, or equivalent.

6. The tapper shall consist of a brass gear, 3/8-inch OD x 1/2-inch in length, having 11 teeth, and mounted on the end of a 1/4-inch fiber rod 5-inches in length. Roll tapper against top corner of tube. Hold tube in test socket with light thumb pressure. The rejection level shall be set at the VU meter reading obtained during calibration.

7. The dc resistance in the grid circuit shall not exceed 2.0 ohms. Measure ac emission as the dc component of current in the anode circuit.

8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter F, shall apply.

9. The TUT shall be rotated to find the direction of vibration in the X plane which gives the highest output reading. For holding fixtures, see Drawing 184-JAN.

10. A grid resistor of 0.5 megohm shall be added except when a thyratron-type short indicator is used.

11. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples. Total defectives shall be 2 and 4, respectively.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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